



Toward traceable XCT measurement of AM lattice structures: Uncertainty in calibrated reference object measurement

Maxwell Praniewicz^{a,b,*}, Jason C. Fox^b, Christopher Saldana^a

^a Georgia Institute of Technology, Atlanta, GA, 30318, USA

^b National Institute of Standards and Technology, Gaithersburg, MD, 20899, USA[†]

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ABSTRACT

Lattice structures created via additive manufacturing are of interest in several demanding industries, however the qualification of these components via X-ray computed tomography (XCT) is limited due to traceability requirements. In this work, a novel measurement reference object is presented to study task specific uncertainty in the XCT measurement of a simplified lattice structure. The design of this reference object, the calibration routines, and XCT acquisition are described. The description of uncertainty calculation is presented. The methodology for data sampling, registration, and measurement is detailed. The measurement results are tested for normality and show variations in the underlying distribution for different measurands. A discussion on uncertainty variation and normality of measurements is then presented. Variations in measurement bias and uncertainty are analyzed across various features of the reference object. It is shown that the measurement uncertainty and bias are not consistent across pin diameter measurement, indicating the need for similarity between the reference object and subsequent component measurements.

1. Introduction

In additive manufacturing (AM), a component is created in a layer-wise manner by sequentially fusing material feedstock using an energy source. This technique allows for the creation of previously unrealizable features—such as complex mold cooling channels, topology optimized structures, and internal features—prompting an entire field of research specifically focused on designing components to maximize the potential of AM [1]. A lattice structure, defined as a “geometric arrangement composed of connective links between vertices (points) creating a functional structure” [2], is one of the most heavily investigated AM-enabled structures and varies significantly in its design and application. The use of AM lattices in orthopedic implants has been shown to significantly improve bone integration and optimize stress shielding at the implant zone [3]. The number of AM implants is expected to grow at a compound annual growth rate (CAGR) of 22.5% from 2020 to 2026 [4]. AM has significant impact in the aerospace industry, with applications ranging from component manufacture using laser powder bed fusion (LPBF) to the repair of high value components using directed

energy deposition (DED) [5]. Lattice structures specifically have been investigated to lightweight spacecraft and commercial aircraft [6]. However, strict component quality requirements imposed by aircraft manufacturers and government regulatory bodies currently limit the use of AM components within these applications. The geometric intricacy of parts and the complexity of the manufacturing process itself make component qualification a non-trivial task.

X-ray computed tomography (XCT) is an imaging technique that utilizes the penetrating properties of X-rays to create a digital reconstruction of a scanned workpiece in terms of a three-dimensional attenuation map comprising volumetric picture elements, or voxels [7]. The volumetric image can be converted to a surface model by applying an attenuation threshold, enabling the measurement of otherwise inaccessible internal features and making XCT a preferred method for inspecting AM components [8]. Internal geometries such as cooling channels, cavities for light-weighting, or embedded dissimilar materials can be analyzed without the need for traditional destructive testing. Tightly compacted lattice structures or complex freeform components can often be analyzed in a single XCT acquisition, as opposed to

* Corresponding author. National Institute of Standards and Technology, Gaithersburg, MD, 20899, USA.

E-mail address: maxwell.praniewicz@nist.gov (M. Praniewicz).

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several disparate measurement processes using other measurement techniques. XCT also enables the non-destructive evaluation of porosity; one of the most common defects in AM parts. Other applications of XCT in the characterization of AM components can be found in the review by Thompson et al. [9].

The geometric qualification of lattice structures using XCT has been investigated previously in the literature to analyze mechanical performance and failure criteria [10–13]. While the use of XCT to characterize AM components has been extremely promising, there is one factor that currently limits its use as a metrology tool: the absence of measurement traceability to the International System of Units (SI). In the medical device and aerospace industries, SI traceability is a requirement for certification of a manufacturer's quality management system [14–16]. The traceable assessment of XCT system performance by manufacturers is progressing towards standardization [17,18], though these standards prescribe the measurement of idealized reference objects. The assessment of task-specific measurement uncertainty, particularly for more complex workpieces, is extremely limited, particularly due to the vast number of factors that influence a single scan [19]. Even the assessment of the uncertainty in one of these factors, i.e. the surface determination, has proven to be non-trivial [20]. Furthermore, it has been shown that, for a given task, it is also a challenge for users to perform measurements with errors within their system's stated specifications [21]. The most commonly implemented methodology to determine task-specific measurement uncertainty in XCT measurements is defined in VDI/VDE 2630 Part 2.1, commonly known as the substitution method [22]. This method relies on multiple CT measurements of a separate, calibrated reference object that must meet strict similarity conditions between reference object and test workpiece. This method has previously been applied to the XCT measurement of plastic and metallic components produced via more traditional manufacturing techniques [23,24]. Many works have examined the use of this methodology and have investigated the use of both corrected and uncorrected approaches for uncertainty estimation in the measurement of components [25]. VDI/VDE 2630–2.1 states that "... the workpieces should be very similar particularly in terms of dimensions, shape (macro and micro shape) and material ...". These conditions are imposed because of non-linearities in X-ray attenuation with respect to object size, shape, and material composition, which can give rise to imaging artifacts, i.e., artificial features in the reconstructed image. As a result, there is a significant interest in the development of reference objects for the assessment of uncertainty in AM-specific XCT measurements. For example, Heřmanek et al. presented a novel reference object and methodology to assess the accuracy of XCT-based porosity measurements [26]. The reference object comprised calibrated machined hemispheres to simulate pores embedded within a workpiece. This work evaluated porosity diameter measurement error as a function of various XCT parameters and measurement algorithms and confirms the high potential of XCT measurement to detect porosity. Zanini et al. used the substitution method to determine the uncertainty in the XCT measurement of an AM lattice component by employing a calibrated set of pins as the reference object [27,28]. However, because of the complex X-ray-workpiece interactions, such as X-ray scatter and beam hardening, there are expected variations between the measurements of the calibrated reference object and the large lattice geometry. The complexities of the AM process also give rise to relatively large variations in surface texture and form errors within lattice geometries [29,30]. It has also been shown that these variations can make be large contributors in the calculation of uncertainty [31]. Thus, it is important to understand the uncertainty in XCT measurement of lattices prior to the inclusion of these factors. The additional contribution by roughness in the measurement of AM lattices can then be accounted for at a later time using previously proposed methods [32]. Thus, further study is needed to evaluate the uncertainty in XCT measurement of a lattice component under conditions (XCT settings, measured features, workpiece construction) that are similar to the evaluation of AM lattice structures. This includes the measurement

of a component of comparable geometric complexity, definition of measurands using product definition standards, and thorough documentation of measurement and evaluation procedures.

In this work, a novel measurement reference object designed to represent a simplified lattice structure is presented to study uncertainty in XCT measurement of lattice structures. Variations in measurement bias and uncertainty are analyzed across various features of the reference object. The design of this reference object, the calibration routines, and XCT acquisition are described. A description of the uncertainty calculation is presented. The methodology for data sampling, registration, measurement, and data analysis is detailed. Measurement results are then reported. A discussion on uncertainty variation is then presented.

2. Theory/Calculations

2.1. Reference object design

The reference object presented in this work can be seen in Fig. 1. To allow for adequate X-ray penetration with lower power XCT systems, the reference object has been designed within the penetration capabilities of an XCT system with a maximum acceleration voltage of 130 kV, as the acceleration voltage is the largest contributing factor to maximum penetration length [33]. The reference object is comprised of two 17-4 PH stainless steel end caps, nine 0.5 mm diameter ASI 52100 steel gage pins, and an acrylic spacer and measures approximately 4 mm in diameter and 6 mm in height. It is designed to represent a section of a larger simple lattice geometry with a repeated pattern of cylindrical struts. The reference object was designed with inspection in mind, compromising geometrical complexity in favor of establishing traceable calibration measurements. While the geometrical complexity may not be equal to that of a "typical" AM lattice structure, the geometry provides a comparable X-ray paths thought out the course of a CT scan. This will lead to similar patterns in X-ray imaging artifacts, such as beam hardening and X-ray scatter, experienced in an AM lattice structure. As this design is limited in size, it is the first step towards evaluating a more complex lattice structure. The end caps provide a datum reference frame for measurements while the acrylic spacer protects the nine pins and separates the two caps. Nine clearance holes in the acrylic component and

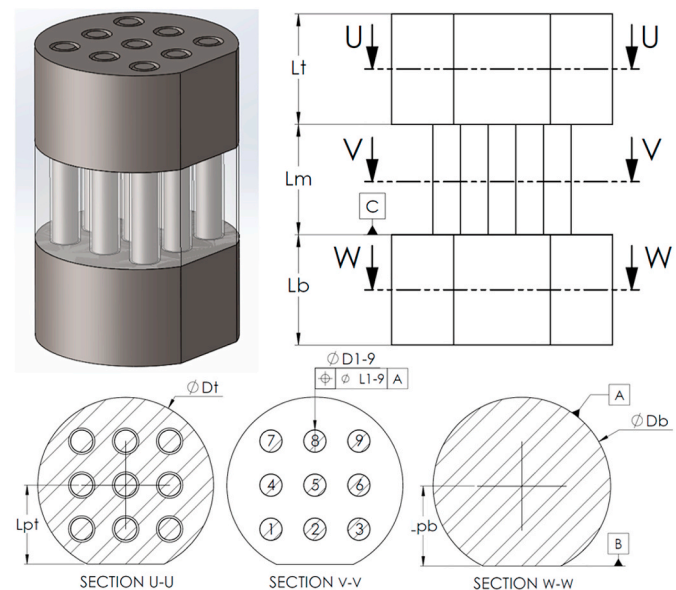


Fig. 1. Reference object design; 3D view and evaluated measurements (shown without acrylic piece). Subscripts "b" and "t" represent measurements for the bottom and top endcap respectively. Section V-V shows the numbering convention of the pins examined in this study.

the top end cap allow the components to be integrated into the assembly. The nine gage pins are fixed into blind holes within the bottom end cap. The end caps were machined on a micro-machining center while the pins were purchased from a supplier. These materials were chosen to minimize differences in X-ray transmission and were calculated to have a transmission difference of only 2% at the target energy using values from the NIST Photon Cross Sections Database (XCOM) data base, the Lambert-Beer law, and the maximum penetration length of 4 mm [34]. Fig. 1 also details the measurements analyzed in this study. The end caps are intended to represent larger geometric features that can exist alongside or bound a lattice within an industrial component. Measurements on each end cap include the cap diameter D , the length L , and the distance of the alignment plane (datum B) to the axis of the primary datum cylinder (datum A) L_p . Measurements of the pins are intended to represent measurements of individual struts of a lattice structure and include the diameter $D_{i=1,2,\dots,9}$ and location relative to the defined datum structure $L_{i=1,2,\dots,9}$ for each pin. A pin location measurement is defined as the linear distance between the cylinder axis of the pin and the cylinder axis of datum at a height of 1.0 mm above datum C in a plane parallel to the axis of datum A. While there are differences between the presented reference object and an AM lattice, primarily in geometrical complexity and surface roughness, it will allow for uncertainty in the XCT system and measurement process to be captured without the uncertainties associated with these geometric and surface complexities. This will allow for future study to examine the influence of AM specific processing parameters on the measurement uncertainty in XCT.

2.2. Uncertainty calculations

The uncertainty of XCT measurements were determined using a modified version of the method described in VDI/VDE 2630 Part 2.1 [22]. According to Eq. (1), the expanded uncertainty of the measurement process, U_{MP} , is given by the summation in quadrature of several factors, which is then multiplied by a coverage factor, k . These factors are the standard uncertainty associated with the calibration of the calibrated workpiece, u_{cal} , the standard uncertainty associated with the measurement procedure, u_p , the standard uncertainty associated with the systematic error of the process, u_b , and the standard uncertainty associated with material and manufacturing variations, u_w .

$$U_{MP} = k \sqrt{u_{cal}^2 + u_p^2 + u_b^2 + u_w^2} \quad (1)$$

A coverage factor $k = 2$ provides the 95.45% confidence interval in the estimated uncertainty, assuming a Gaussian distribution. The goal of this work is to investigate the effect of lattice geometrical complexity on the uncertainty in XCT measurement. Thus, only the uncertainty contributors which are influenced by measurements of a calibrated workpiece can be examined. The effects related to AM specific manufacturing errors, such as rough surface textures and form errors, cannot be determined in the present work. Therefore, only the first three terms in Eq. (1) are assessed. Specifically referring to VDI 2630 Part 2.1 Section 4.2.2, our work encompasses steps 1 and 2 of the complete process for the measurement of workpieces. Therefore, the expanded uncertainty expressed for measurements in this work, U_i , can be seen as a combination of the first three contributors, shown in Eq. (2), and will provide information to the calculation of U_{MP} for the measurement of similar AM lattice structures in future work.

$$U_i = k \sqrt{u_{cal}^2 + u_p^2 + u_b^2} \quad (2)$$

The standard states that u_b must at least contain the standard uncertainty in the measurement of the linear coefficient of expansion of the calibrated workpiece. The recommended calculation for u_b is presented in Eq. (3), where T is the average temperature of the reference object, u_α is the uncertainty in the coefficient of thermal expansion of the reference object, and l is the measured dimension.

$$u_b = (T - 20^\circ\text{C})u_\alpha l \quad (3)$$

However, if measurements are performed at non-standard temperatures and not in a controlled environment, such as on a shop floor, additional terms should be accounted for, as described in Ref. [35]. Furthermore, since a single XCT scan can take place over the period of an hour, there may be environmental fluctuations throughout the scan. Therefore, this work proposes to modify Eq. (3) to Eq. (4) to account for these factors.

$$u_b = \sqrt{((T - 20^\circ\text{C})u_\alpha l)^2 + (a l u_T)^2 + \left(\frac{s_b}{\sqrt{n}}\right)^2} \quad (4)$$

In this equation s_b is the standard deviation of the calculated bias errors and n is the number of measurements. The uncertainty associated with the average temperature measurement, u_T , is calculated using Eq. (5).

$$u_T = \sqrt{u_{Td}^2 + u_{Tm}^2} \quad (5)$$

The temperature uncertainty can come from two sources: standard uncertainty in the temperature measurement device, u_{Td} , and the standard deviation of the temperature throughout the measurement process u_{Tm} .

It is assumed, for the results presented, that the bias can and will be corrected. Therefore, the bias result are presented separate from the calculated uncertainties and were calculated using Eq. (6). The bias, b , is the result of subtracting the calibrated value of a measurand, y_{cal} , from the mean of repeated measurements, $\bar{y}$.

$$b = \bar{y} - y_{cal} \quad (6)$$

Using this method, a final measurement result, X , would reported as in Eq. (7) where the measurement would be compensated by the bias.

$$Y = y - b \pm U_i \quad (7)$$

2.3. Calibration

The various measurands on the reference object were calibrated using a Zeiss Micura coordinate measurement machine (CMM) with a maximum permissible error of length measurement ($E_{0,MPE}$) of $(0.8 + L/400)$ μm , where L is the measured length in mm, according to ISO 10360-2 [36]. To assess all features, calibration was completed in multiple steps. In the first setup, the end caps were individually measured to densely sample the outer diameter, top plane, and the alignment plane. This setup allowed for the calibration of D_t and L_{pt} for the top end cap and D_b and L_{pb} for the bottom end cap. The points acquired in this setup can be seen in Fig. 2 (a)–(c). In a second setup, the components were individually fixtured in a custom v-block, which allowed for the measurement of the top and bottom plane. This setup

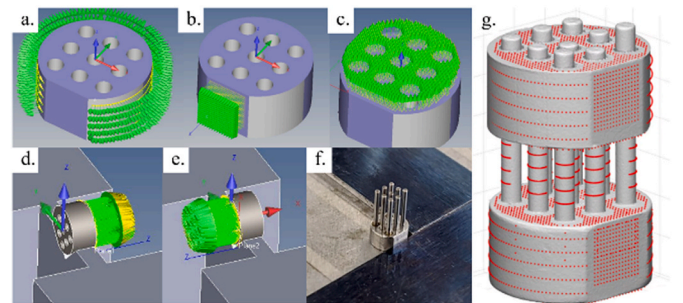


Fig. 2. CMM measurements: (a) Outer diameter measurement (b) Alignment plane measurement (c) Top plane measurement in setup one (d) Top plane measurement in setup two (e) Bottom plane measurement (f) Pin location measurement (g) Compiled CMM points.

allowed for the measurement of L_t on the top end cap, L_m on the middle acrylic piece, and L_b on the bottom end cap. The points acquired in this measurement setup can be seen in Fig. 2 (d),(e). In the final setup, the pins were inserted and fixed into the base using hard setting epoxy. After the epoxy had cured, the location of the pins was measured relative to the defined datum structure shown in Fig. 1. All measurements were performed with a 50 mN probing force, a 0.5 mm probe tip diameter, and active thermal compensation using a coefficient of thermal expansion (CTE) of $10.8 \mu\text{m}/\text{m}\cdot\text{K}$ for the metallic components and a CTE of $85 \mu\text{m}/\text{m}\cdot\text{K}$ for the acrylic spacer. The uncertainty in the CMM measurements was calculated similar to the procedure described in Shah et al. [37], which utilized the methodology described in ISO 14253-2 [38]. Since thermal compensation was active throughout the course of the measurements, the uncertainty in temperature fluctuations was not included. Repeated measurements of the reference object features also showed negligible variations. Therefore, measurement repeatability was also not included in the calibration uncertainty. Thus, the uncertainty in a calibration of an individual feature was defined as $u_{cal} = \frac{E_{0,MPE}(L)}{2}$, where L is the largest distance between measured points. The diameter of the pins was not calibrated during the CMM measurements and was substituted with the certificate of conformance of the class Y bilateral gage pins per ANSI/ASME B89.1.5 1998 provided by the manufacturer [39]. Since no knowledge of the manufacturing process is available to the authors, a uniform distribution across the conformance tolerance range was assumed to calculate the standard uncertainty in calibration, yielding a standard uncertainty of $0.5 \mu\text{m}$. The results from the calibration measurements and their associated uncertainty are listed in the Appendix. After all points had been captured from the individual components, they were imported into MATLAB for further processing. The pin measurements and bottom end cap measurements acquired in each component orientation were registered within the same coordinate system using the measured datum features. The data points for the top end cap were placed at a nominal distance L_m above the bottom endcap. The final registration of the measured CMM points can be seen in Fig. 2 (g).

2.4. XCT measurement/Analyses

Thirty repeat XCT acquisitions of the reference object were performed using a Zeiss Metrotom 800 XCT system, which has a manufacturer stated sphere distance maximum permissible error (SD_{MPE}) of $[0.0045 + L/100] \text{ mm}$ according to VDI/VDE 2630 Part 1.3 [40]; the measurement parameters were identical in all acquisitions. During all scans, the temperature within the system was monitored using a wireless environmental sensor, which sampled the temperature once per minute and was confirmed to have a standard deviation of 0.1°C during an internal gage repeatability and reproducibility study. Prior to the replicate scans, all manufacturer defined characterization and correction routines were performed. These included the characterization and correction for system geometry, manipulator axis motions, and detector response. Using a batch measurement, all scans were performed in sequence with the same starting position. Each scan lasted approximately 65 minutes, meaning the system ran continuously for approximately 35 hours. The parameters used in the individual scans can be found in Table 1. In addition to the listed parameters, machine specific settings for focal spot control, image correction, and ring artifact compensation were enabled.

The acquisition setup, including the enclosure of the reference object in an acrylic stand, can be seen in Fig. 3 (a),(b). An example projection can be seen in Fig. 3 (d). A Feldkamp-type Filtered Back Projection (FBP) algorithm [41], which utilized a Shepp-Logan filter, provided by the CT system manufacturer was applied to the acquired projections. No beam-hardening correction techniques were utilized. A sample histogram of the reconstruction can be seen in Fig. 3 (c). The reconstructed volume was then imported into VGStudio MAX 3.4 [42]. The component

Table 1

XCT parameter scan parameters.

Parameter	Value
Voltage	130 kV
Current	61 μA
Number of Projections	900
Images Averaged Per Projection	2
Integration Time	1 s
Source to Detector Distance	803.359 mm
Source to Object Distance	60.606 mm
Voxel Size (VS)	$9.58 \mu\text{m} \times 9.58 \mu\text{m} \times 9.58 \mu\text{m}$
Physical Filter	0.5 mm Cu
Focal Spot Size	8 μm

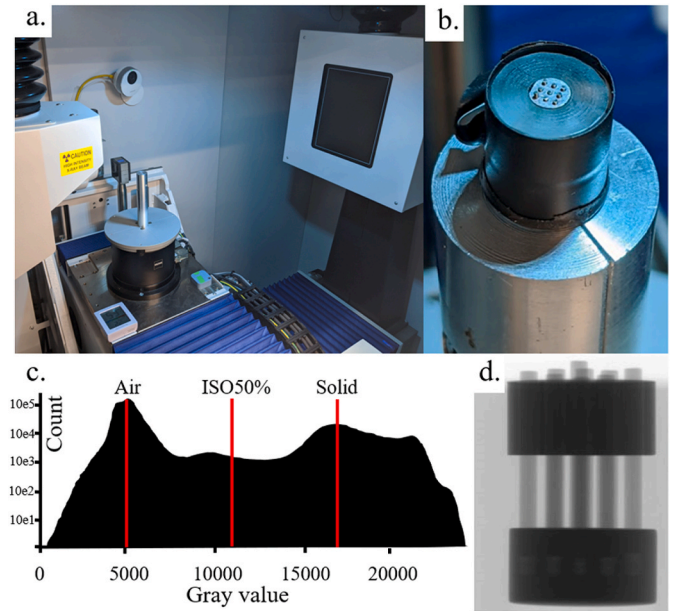


Fig. 3. XCT measurement: (a) Measurement Setup within the XCT system (b) Reference object within acrylic enclosure (c) Example histogram (d) Example projection.

surface was initially determined using ISO50 thresholding, followed by an advanced thresholding using an 8 voxel search distance and iterative surface determination selected. An example thresholded volume is shown in Fig. 4 (a).

The extracted surface models were initially registered using the datum scheme shown in Fig. 1 by fitting geometry elements to the datum surface in VGStudio MAX using least-squares. The registered surface models were then converted to a stereolithography (STL) file and exported using a meshing tolerance of $1 \mu\text{m}$ and a mesh resampling of 5

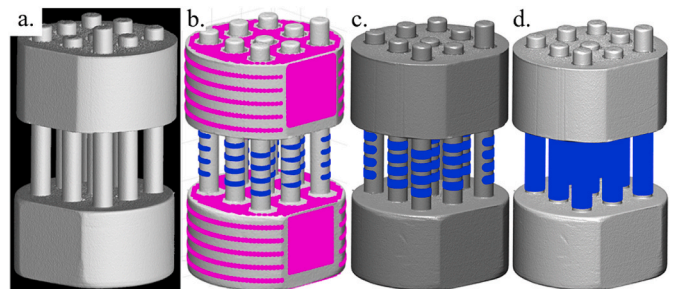


Fig. 4. Representative XCT data (a) Thresholded volume (b) Points sampled for feature measurement (c) Pin sampling based on CMM locations (d) Pin sampling based on entire pin surface.

μm . To minimize variation in measurement assessment between the XCT data and the CMM data, the sampling strategy described in Ref. [43] was utilized and each mesh was registered to the CMM data using the defined datum schema. However, the pin diameter measurements utilized sampling the entire surface of the pin, as the calibration of this value is not related to the CMM points. This sampling can be seen in Fig. 4 (d). These two different sampling strategies, shown in Fig. 4 (c) and (d), were also compared in the pin location measurements and were designated as $L_{i=1,2,\dots,9}(\text{CMM})$ and $L_{i=1,2,\dots,9}$, respectively. Features were assessed by using least-squares fitting algorithms constructed as described in Ref. [44]. An example of the sampled XCT measurement points for the bottom and top end caps can be seen in Fig. 4 (b). The accuracy of least-squares cylinder and plane fitting algorithms used was verified using reference datasets from the NIST algorithm testing service [45]. For each feature, the standard deviation of the thirty repeated XCT measurements was used as u_p , the standard uncertainty associated with the measurement procedure, defined in Eq. (8).

$$u_p = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

To compensate for measurement at non-standard temperatures, the bias values of feature measurements within an individual scan were also compensated by the mean temperature throughout the course of the individual scan. The calibration values, measurement data from all thirty scans, and temperature readings were combined to form an uncertainty statement for each feature.

2.5. Measurement normality analysis

The underlying distribution in repeat measurements is often assumed to be Gaussian because a repeatable process is often expected to be symmetric about the mean. To test for normality of the measured XCT data, a Shapiro-Wilk test with a significant level of 0.05 was performed on the data gathered from each feature. The Shapiro-Wilk test was chosen as the test for normality because of its statistical power compared to other methods [46].

VDI/VDE 2630 Part 2.1 defines u_p as the standard deviation of more than 20 replicate measurements, as in Eq. (8). If the measurement data presented in this work has been shown not to follow a Gaussian distribution and has been shown to have significant outliers, it may be more appropriate to use a different measure of dispersion to define u_p . The average absolute deviation (AAD), defined in Eq. (9) and the median absolute deviation (MAD), defined in Eq. (10) are other measures that can be used to characterize the dispersion of data. These two metrics are less affected by outliers, which can significantly skew the calculation of standard deviation.

$$AAD = \frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}| \quad (9)$$

$$MAD = \text{median}[|y_i - \bar{y}|] \quad (10)$$

These two measures of dispersion can be compared to the standard deviation if scaled by the appropriate factor. For normally distributed data, these factors are 1.2533 and 1.4628 for the AAD and MAD, respectively [47]. While the authors acknowledge that the data presented is not normally distributed, these scaling factors were used to allow relative comparison to standard uncertainty using the standard deviation. If non-normal, calculation of measurement uncertainty will be repeated using two different metrics:

$$u_p = 1.2533 \cdot AAD \quad (11)$$

$$u_p = 1.4628 \cdot MAD \quad (12)$$

3. Results

3.1. Pin diameters

Each set of diameter measurements $D_{i=1,2,\dots,9}$ was examined using the Shapiro-Wilk test and accepted the null hypothesis of normality with calculated P-Values greater than 0.1. The bias from CMM measurements and uncertainty estimates in the XCT measurement of pin diameters are plotted in Fig. 5 (a). The red points correspond to the bias in the mean of thirty repeated XCT acquisitions, while the associated plotted error bars correspond to the expanded uncertainty, U_t , for each feature. The largest bias observed for these measurements was $0.4 \mu\text{m}$ (pin 3), which is less than 5% of the $9.58 \mu\text{m}$ voxel size (VS). The average bias over all pins ($0.0 \mu\text{m}$) and the variation in bias among pins is minimal and indicates very consistent agreement with the calibrated reference. The spatial variations in bias are illustrated in Fig. 5 (b). Pin 5 is observed to have the highest bias value, which could possibly be caused by cupping artifacts as a result of beam hardening. As noted in Section 2.4, the reference object was not moved between measurements. Thus, it is possible that the spatial variation is simply due to the partial volume effect [33]. All bias values are also tabulated in Table 2. The average U_t over all diameter measurements is $1.0 \mu\text{m}$ with minimal variation. These results suggest a remarkable repeatability in the measurement of a metal workpiece, given the pronounced effects of X-ray scatter and beam hardening.

For comparability to other studies which have investigated the measurement uncertainty of XCT, the ratio of VS to measurement bias was calculated in a manner similar to Carmignato et al. [21]. Table 2 displays the results of this calculation. While the results presented in Ref. [21] are for a completely different measurement task, they provide a relative metric of quality to assess the results. The average of the ratio of VS to measurement bias for all pin measurements was 72.8, showing excellent performance in diameter assessment. The ratio of VS to the calculated U_t for the diameter measurements is also presented in Table 2, and indicates that the uncertainty in strut diameter measurements with XCT is significantly smaller than the voxel size.

To investigate the spatial variation in bias and U_t between pins in the reference object, pin measurements were combined to assess the uncertainty associated with the measurement of different groups of pins. In the first group, the measurements of the corner pins were combined, as these pins were at similar radial positions from the center of the rotational axis during the scan. Similar patterns in penetration length for these four pins would lead to similarities in attenuation and the potential for artifacts relating to beam-hardening and scatter. Thus, they were grouped to form a set of 120 diameter measurements D (1,3,7,9). The side pins were similarly grouped to form D (2,4,6,8). To assess the uncertainty in the measurement of a given pin diameter in the reference object, all measurements were grouped to form a set of 270 measurements D (1–9).

In the Shapiro-Wilk test, all groups do not appear to follow a Gaussian distribution and reject the null hypothesis of normality with calculated P-Values less than 0.01. Fig. 6 displays the results from the grouped measurements. Since the tests for normality reject the null hypothesis, the results in Fig. 6 (a) are shown with U_t calculated using the three previously defined measures of dispersion. The uncertainty values reported in Fig. 6 (c) and the U_t values reported in the text were calculated using the standard deviation as u_p for comparison to the results reported in Fig. 5. One can see minimal differences between the three different groups. Despite the non-normal data sets, marginal differences are observed between the different measures of spread ($\sim 20 \text{ nm}$) for all three groupings. The spatial plots in Fig. 6 (b) and (c) display a variation in average bias and uncertainty, indicating that the corner pins, which experience the largest variations in total penetration distance throughout the acquisition of projections, have a lower average bias. Spatial variation can also be observed in the U_t , indicating a marginally higher uncertainty in D (2,4,6,8). The D (1–9) results

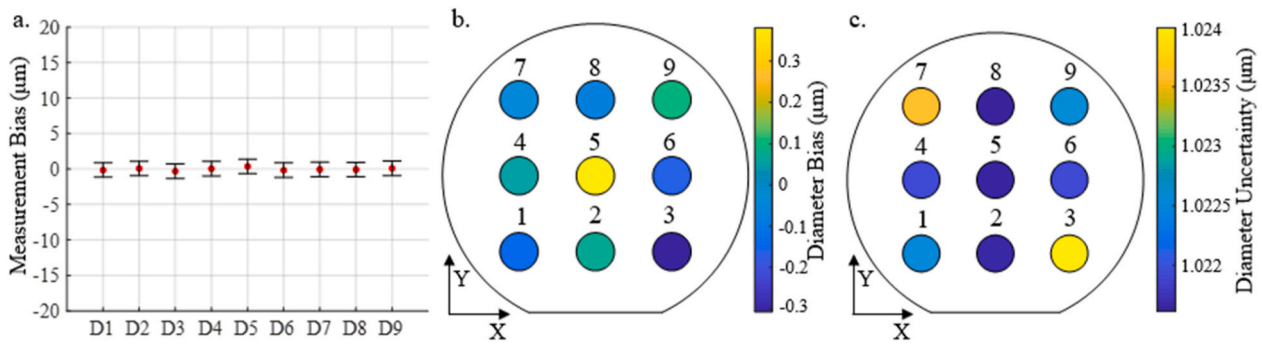


Fig. 5. Results of pin diameter XCT measurement. (a) The red dots indicate the measured bias of the mean over thirty repeat acquisitions from the CMM calibrated value. Error bars indicate the U_t in the measurement process. (b) Diagram illustrating the spatial distribution of pin diameter bias. (c) Diagram illustrating the spatial distribution of pin diameter uncertainty, U_t . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 2
Pin diameter measurement results.

Measurement	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9	Average
b (μm)	−0.1	0.1	−0.3	0.1	0.4	−0.2	0.0	−0.1	0.1	0.0
U_t (μm)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
VS/b	95.8	95.8	31.9	95.8	24.0	47.9	∞	95.8	95.8	72.8
VS/U_t	9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4	9.4

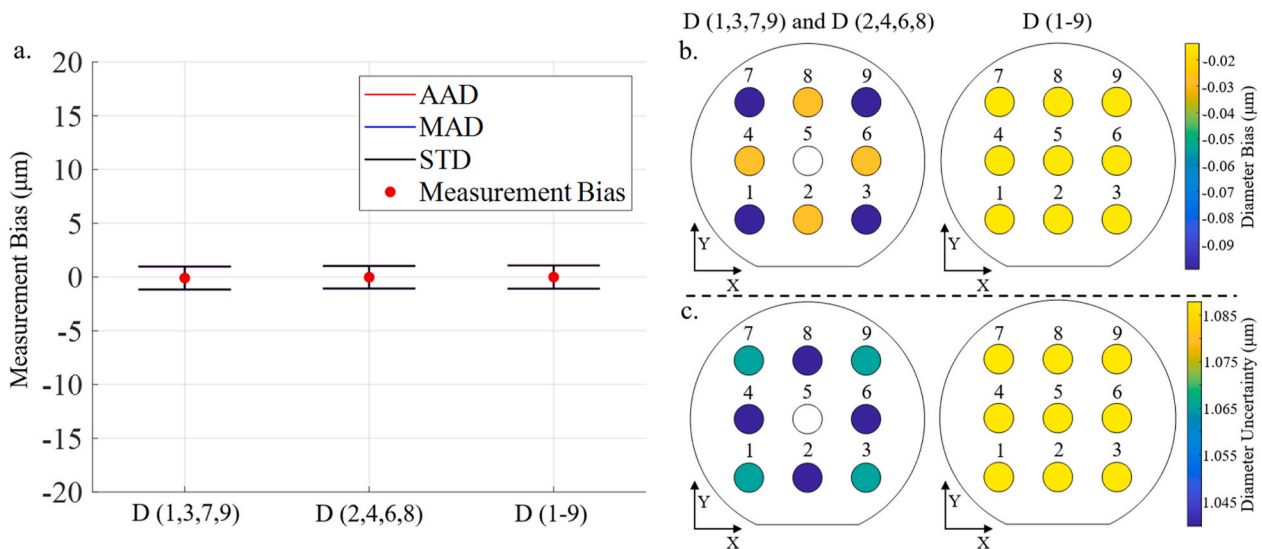


Fig. 6. Results of grouped pin diameter XCT measurement. (a) Measurement bias for the three different pin groups. Error bars indicate the U_t in the measurement process calculated with different measures of dispersion. (b) Diagram illustrating the spatial distribution of pin diameter bias. Left spatial plot displays the corner and side pins, while the right spatial plot displays the bias if all pins are grouped. (c) Diagram illustrating the spatial distribution of pin diameter U_t calculated using the standard deviation as u_p . Left spatial plot displays the corner and side pins, while the right spatial plot displays the bias if all pins are grouped.

indicate the U_t calculated using the standard deviation is the most conservative.

3.2. Pin locations

The set of pin location measurements using sampling around the CMM points was examined first. The measurements $L_{i=1,2,\dots,9}(CMM)$ were examined using the Shapiro-Wilk test and all measurement sets were found to reject the null hypothesis of normality with calculated P-Values less than 0.01. Fig. 7 displays the results from the XCT measurement and uncertainty analysis for pin location measurements using the three different measures of spread. Significant differences in both bias and measurement uncertainty can be observed among the various

pins. The spatial plots of the U_t in Fig. 7 (c) are calculated using the standard deviation. The bias values range between +13.6 μm and −16.9 μm, corresponding to pins 7 and 3, respectively. The average observed bias is −1.1 μm. While this indicates that on average the pins are closer to the A axis than in the calibration, no apparent trends are observed in location measurement. While there initially appears to be a trend indicating a shift in +Y when assessing the corner pins (pins 7 and 9 with positive bias and pins 1 and 3 with negative bias), the side pins (2,4,6,8) do not agree with this. The average U_t of location measurement is −2.0 μm, double the average U_t in diameter measurements. One can also see differences in the U_t using the different measures of spread. For the $L_{i=1,2,\dots,9}(CMM)$ measurements, one can observe that the standard deviation provides the most conservative estimate of uncertainty. It is not

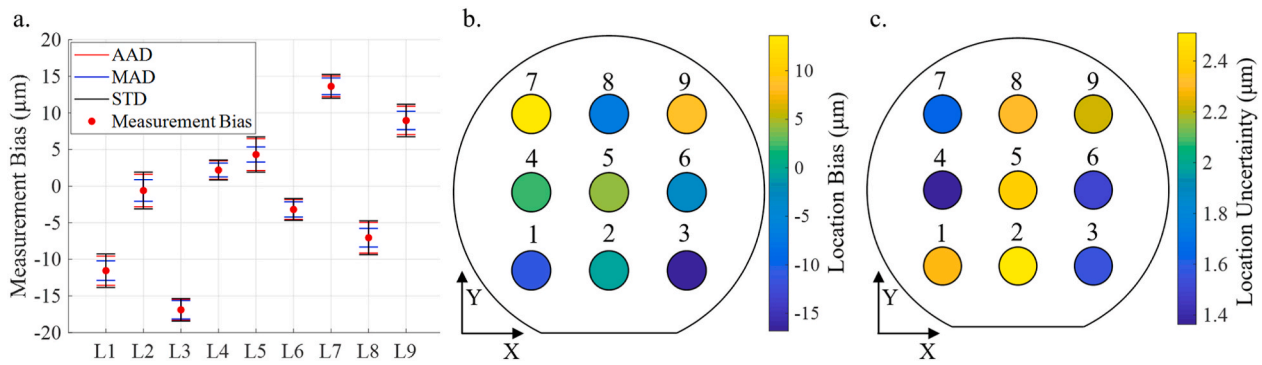


Fig. 7. Results of pin location XCT measurement using CMM sampling locations. (a) The red dots indicate the measured bias of the mean over thirty repeat acquisitions from the CMM calibrated value. Error bars indicate the U_t in the measurement process using the three different measures of spread (Standard deviation in black, MAD in blue, and AAD in red). (b) Diagram illustrating the spatial distribution of pin location bias. (c) Diagram illustrating the spatial distribution of pin location U_t calculated using the standard deviation as u_p . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

immediately apparent to the authors why the bias values assessed for location were much larger than the diameter measurements, as the bias in non-threshold dependent measurements is often lower [48].

To examine the possibility of spatial sampling influencing the location measurement, the same analysis was performed on measurements $L_{i=1,2,\dots,9}$ as in $L_{i=1,2,\dots,9}(\text{CMM})$, since it has been shown that location dependent form errors on individual features can appear as a result of non-uniform projection length throughout projection acquisition [49]. Fig. 8 displays the measurement results. The spread of the bias values is seen to decrease in these measurements, ranging between +10.3 μm and −12.8 μm, corresponding to pins 7 and 8, respectively. However, the magnitude of the bias is not seen to decrease for all pins. The uncertainty of all location measurements is seen to increase significantly, with the average in Fig. 8 (c) almost three times that observed in Fig. 7 (c). The U_t is seen to increase from 5.6 μm to 8.4 μm between using the standard deviation and the MAD in pin 8, indicating that the chosen measure of spread can greatly impact the final reported uncertainty. These results expose the impact that data sampling can have on both the bias and uncertainty in XCT measurements.

To determine if the bias in pin location measurement was a result of unintended pin deformation when adding the acrylic spacer and top end cap to the assembly, the form of the measured pins was also assessed on the $L_{i=1,2,\dots,9}$ data sets. Cylindrical form was determined for a given pin by calculating the deviations between the sampled XCT data points and the fit cylinder and subtracting the minimum deviation from the maximum deviation. Table 3 shows the results of the form measurement of the nine pins over the thirty repeated XCT acquisitions. Form was

assessed on individual point deviations, which can be influenced by the partial volume effect, local thresholding, and the generation of the triangulated surface from the surface model. Considering a VS of 9.58 μm and a meshing tolerance of 1.0 μm, the calculated average form error of 6.8 μm for all pins is acceptable, but well above the B89.1.5 “Limit for roundness, taper, or straightness” of 0.89 μm per the certificate of conformance [39]. The ratio of VS to form measurement is also presented. The calculated average ratio of 1.4 is above the median of the results found in Ref. [21], indicating acceptable form measurement. Since the average form error calculated for each individual pin was less than a single size, there is not enough evidence to conclude that any of the pins was deformed within the assembly process.

TVDI/VDE 2630 Part 2.1 follows the ISO GUM [50] and recommends that all systematic errors observed in the measurement of reference objects be corrected in subsequent measurement of workpieces. In this work, we assume that the bias observed can be corrected. We base this assumption on the high repeatability of the measurement results and the quality of the reference object calibration. However, when further measurements of workpieces are performed (e.g., measurements of an AM lattice structure of similar design), there is potential that this assumption may not hold true due to influences from the workpiece geometry. If this assumption does not hold true, then the bias should be incorporated into the uncertainty statement. VDI/VDE 2630 recommends to add the bias in quadrature with the other uncertainty standard uncertainties. However, this method has been shown to overestimate the uncertainty compared to other methods [51]. This is similar to the recommendations and observations reported in Ref. [25].

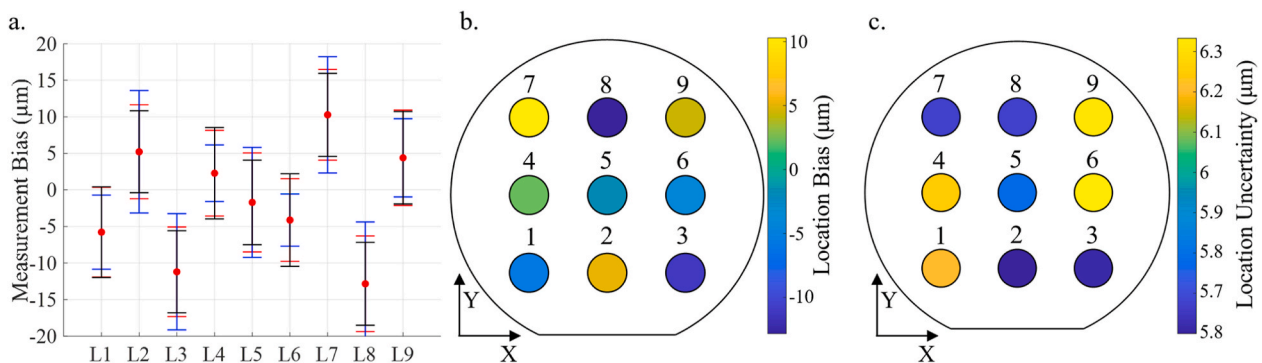


Fig. 8. Results of pin location XCT measurement using sampling over entire pin surface. (a) The red dots indicate the measured bias of the mean over thirty repeat acquisitions from the CMM calibrated value. Error bars indicate the U_t in the measurement process using the three different measures of spread (Standard deviation in black, MAD in blue, and AAD in red). (b) Diagram illustrating the spatial distribution of pin location bias. (c) Diagram illustrating the spatial distribution of pin location U_t calculated using the standard deviation as u_p . (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 3
Pin form measurements.

Measurement	Pin 1	Pin 2	Pin 3	Pin 4	Pin 5	Pin 6	Pin 7	Pin 8	Pin 9	Average
Form (μm)	6.0	7.2	8.8	6.0	5.7	7.0	8.5	6.1	6.2	6.8
VS/Form	1.6	1.3	1.1	1.6	1.7	1.4	1.1	1.6	1.4	1.4

3.3. End caps

Fig. 9 displays the results from the XCT measurement and uncertainty analysis for the end cap features. One can see relatively high bias is observed in the measurement of L_b and L_t , with values of $8.5 \mu\text{m}$ and $3.4 \mu\text{m}$, respectively. This is likely due to presence of cone beam artifacts, as these surfaces were perpendicular to the stage axis of rotation. Distortions along these edges would extend the thresholded surface, skewing the least-squares fit plane farther from the actual surface and creating a positive bias. These distortions can be observed in Fig. 4 (a). Despite these effects, the U_t of L_b and L_t are $0.9 \mu\text{m}$ and $0.8 \mu\text{m}$, respectively, suggesting that the measurement process itself is repeatable. The remaining end cap features exhibited similar measurement uncertainty to those observed in the pin diameter measurements.

Despite the complexity and numerous steps associated with the XCT measurement process— including projection acquisition, reconstruction, thresholding, data sampling, and feature fitting— the presented results indicate high repeatability.

4. Analysis/discussion

While it is important that environmental monitoring should be captured throughout the course of XCT acquisitions, the data presented in this work shows that, in this case, it does not contribute significantly to U_t . On average, the temperature over the course of a scan was found to fluctuate by no more than 0.08°C , which is within the uncertainty of the environmental sensor itself. The mean temperature for an individual scan was found to fluctuate 0.8°C between the thirty scans, for which the overall mean temperature was 22.4°C , thus requiring thermal compensation to the standard temperature of 20°C . Fig. 10 displays the percentage contribution of each component in Eq. (1) to the combined standard uncertainty for the grouped pin diameter measurements, where the percentage contribution is calculated per Eq. (13), in which $\frac{u_i}{u_c} = u_i$ is the combined standard uncertainty and u is a given component of the standard uncertainty.

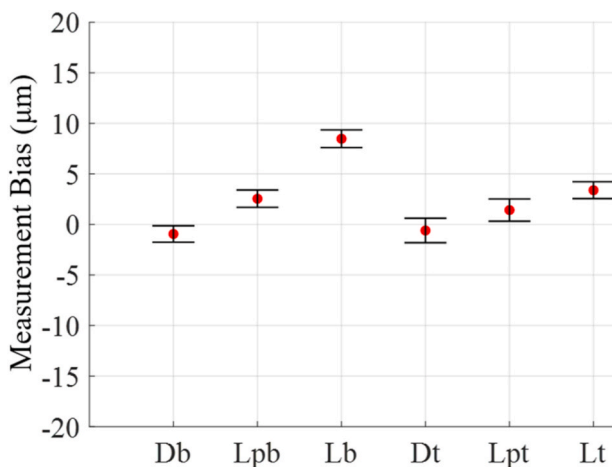


Fig. 9. Results of end cap XCT measurement. The red dots indicate the measured bias from the calibrated value. Error bars indicate the U_t in the measurement process. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

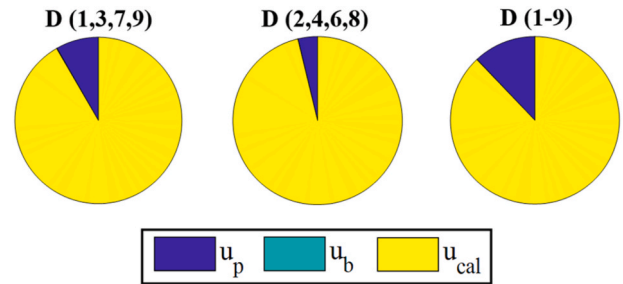


Fig. 10. Contribution of individual components to the combined standard uncertainty in grouped pin diameter measurements.

$$\% \text{ Contribution} = \frac{u_i^2}{u_c^2} 100\% \quad (13)$$

Table 4 displays the values for each of the individual uncertainty components in the grouped pin diameter measurement. The standard uncertainty of bias, u_b , contributes minimally to the total uncertainty. While minimal in these results, u_b is still an important factor as it should always be considered when measuring at non-standard temperatures. The minimal contribution of u_b is also influenced by the relatively small nominal lengths considered and will increase with l according to Eq. (3). Surprisingly, the dominant factor in the total uncertainty is the standard uncertainty of calibration, u_{cal} . This observation differs from other work in the literature, where the standard uncertainty associated with the measurement procedure, u_p , was often the dominant factor [23,52]. The relatively low u_p observed in this work points to a highly repeatable measurement process for some measurands. This high repeatability could be, at least in part, because the component was not removed from the system and re-fixtured between scans, and therefore remained in the same position within the machine. While the influence of fixture repeatability has been studied for previous measurement and manufacturing operations, its influence on XCT measurements has not yet been thoroughly investigated. This could also lead to the isolation of errors which vary within the volume of the XCT system, which were not thoroughly explored in this study. Another possibility for the relatively large contribution of u_{cal} is the relatively large standard uncertainty of the gauge pins. If the diameters of these pins were to be individually calibrated instead of relying on the certificate of conformance, the relative contribution of u_{cal} would drop significantly [53]. By grouping the pins at locations which should nominally experience the same changes in penetration length throughout the course of the scan, the localized biases can be averaged out to gain a better understanding of the overall/general measurement error throughout the reference object. This analysis results in an increase in the variability of the group, increasing the uncertainty of the group and leading to the highest contribution of u_p in the D (1–9) group. This larger variability can also be

Table 4
Uncertainty component values for the grouped pin diameter measurements (rounded to $0.1 \mu\text{m}$).

	u_p (μm)	u_b (μm)	u_{cal} (μm)	u_t (μm)	b (μm)
D (1, 3, 7, 9)	0.2	0.0	0.5	0.5	−0.1
D (2, 4, 6, 8)	0.1	0.0	0.5	0.5	0.0
D (1 – 9)	0.2	0.0	0.5	0.5	0.0

seen in the D (1,3,7,9) group when compared to the D (2,4,6,8) group.

Fig. 11, which shows the relative uncertainty contributions for the $L_{i=1,2,\dots,9}$ measurements, does not indicate the same trend as the diameter measurements. The largest contributor to uncertainty in pin location measurement is the repeatability of the measurement process u_p , which makes up $\sim 95\%$ of the combined standard uncertainty in all measurements and ranges between $2.7\ \mu\text{m}$ and $3.1\ \mu\text{m}$ throughout the nine measurands. These values are also listed in Table 5. The second largest portion of the combined standard uncertainty is u_b , which makes up $\sim 3\%$ of the combined standard uncertainty, contributing between $0.5\ \mu\text{m}$ and $0.6\ \mu\text{m}$. The high variability in these measurements when compared to the diameter measurements could be attributed to the complexity of the measurement itself. In each measurement, data is first sampled from the base features. Geometric features are then fit to the sampled data and the datum reference frame is constructed. Then, data is sampled from the pins, cylinders are fit to the data, and the location of the cylinder is determined with respect to the datum reference frame. Slight variations in any one of these steps for a given measurement would lead to variations in the final measurement result, compared to simply sampling and fitting for the diameter measurements.

Fig. 12 shows the uncertainty contributions of the individual components in the end cap measurements and Table 6 details the values for these components. The repeatability of the end cap measurement is generally not the major contributing factor in the total uncertainty. The repeatability is much higher for both D_b and Lp_b than it is for D_t and Lp_t . In the end cap diameter measurements, the contribution of u_p on the bottom end cap is $<1\%$ while, in the top end cap, u_p constitutes 53% of the total uncertainty. While these measurements are not directly related to the datum reference frame like the pin location measurements, the differences in uncertainty could be related to changes in data registration. Since the bottom end cap contains the datum surfaces for alignment, its position is likely the most repeatable. If small variations in alignment occurred throughout the thirty measurements, the magnitude of linear deviations would increase as the distance from the datum reference frame origin increased. If variations in alignment occurred, variations in point sampling would be expected using the method

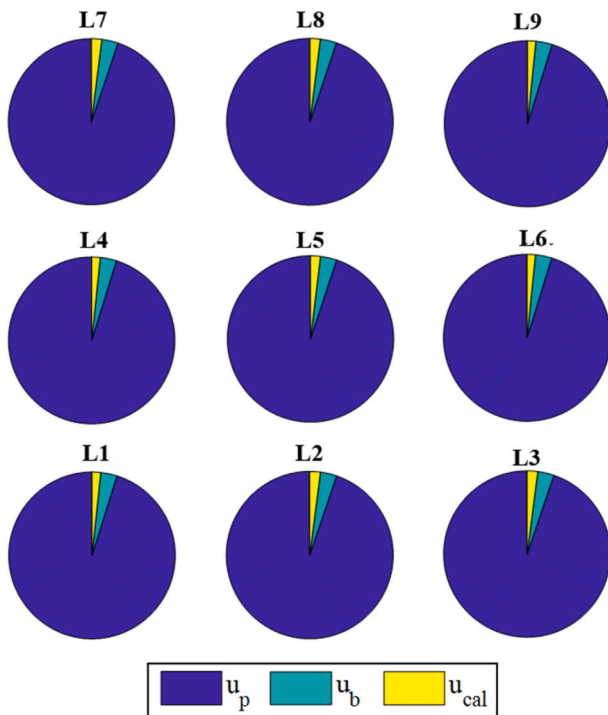


Fig. 11. Contribution of individual components to the combined standard uncertainty in pin location measurements.

Table 5

Uncertainty component values for the $L_{i=1,2,\dots,9}$ measurements (rounded to $0.1\ \mu\text{m}$).

	$u_p\ (\mu\text{m})$	$u_b\ (\mu\text{m})$	$u_{cal}\ (\mu\text{m})$	$u_t\ (\mu\text{m})$	$b\ (\mu\text{m})$
L_1	3.0	0.6	0.4	3.1	-5.8
L_2	2.7	0.5	0.4	2.8	5.2
L_3	2.7	0.5	0.4	2.8	-11.2
L_4	3.0	0.6	0.4	3.1	2.3
L_5	2.8	0.5	0.4	2.9	-1.7
L_6	3.1	0.6	0.4	3.2	-4.1
L_7	2.8	0.5	0.4	2.8	10.3
L_8	2.8	0.5	0.4	2.8	-12.8
L_9	3.1	0.6	0.4	3.2	4.4

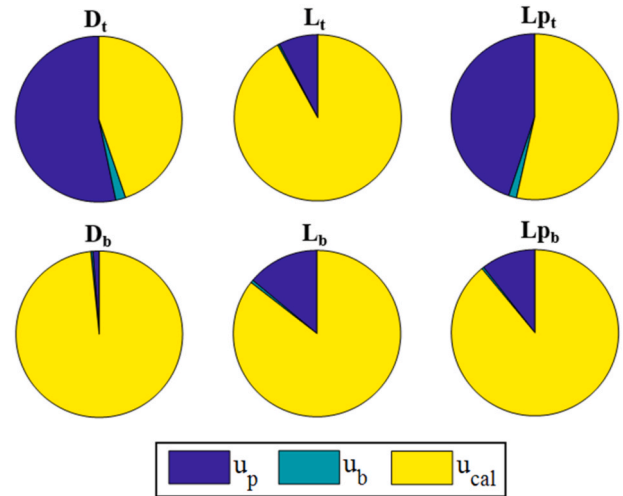


Fig. 12. Contribution of individual components to the combined standard uncertainty in end cap measurements.

Table 6

Uncertainty component values for end cap features (rounded to $0.1\ \mu\text{m}$).

	$u_p\ (\mu\text{m})$	$u_b\ (\mu\text{m})$	$u_{cal}\ (\mu\text{m})$	$u_t\ (\mu\text{m})$	$b\ (\mu\text{m})$
D_t	0.4	0.1	0.4	0.6	-0.7
L_t	0.1	0.0	0.4	0.4	3.4
Lp_t	0.4	0.1	0.4	0.5	1.5
D_b	0.0	0.0	0.4	0.4	-0.9
L_b	0.2	0.0	0.4	0.4	8.5
Lp_b	0.1	0.0	0.4	0.4	2.5

described. While small, these variations in sampling over the determined surface could lead to changes in the measurement repeatability. Thus, increasing u_p for the top end cap features.

5. Conclusion

In this work, a reference object comprising nine cylindrical pins mounted inside an acrylic spacer and between two end caps representing a lattice geometry was presented. The reference object was calibrated using traceable measurement techniques. The calibration results, environmental monitoring, and thirty replicate XCT scans of the reference object were used to determine measurement uncertainty for various features using the substitution method. Analyses of the end cap features showed that even with significantly larger total penetration length, low uncertainty in measurement was achievable. Overall, the results presented showed the viability of the designed reference object for diameter measurement and end cap feature measurement. While the assessment of u_p using the standard deviation of measurement results, on average, provides the most conservative estimate of uncertainty, other measures

of dispersion may provide a more accurate characterization of the standard uncertainty when analyzing non-normally distributed data. As shown in the measurement of pin location, the variations between the two techniques can be significant. This work shows that even in the measurement of a simple, ideal, lattice reference object, significant challenges must be overcome to provide a statement of measurement uncertainty in XCT measurement and highlights the significant differences in measurement uncertainty across different measurands. This information is critical to those who are creating design and production requirements for lattice structures for a variety of applications. This information will allow designers to make informed choices in geometric tolerance specification. Without knowledge of measurement uncertainty, these specifications cannot be met or defined. It has also been shown in this work that ambiguous description of data sampling can lead to significant variations in the measurement result. In the case of AM lattice structures, the process is further complicated by surface roughness and form errors which can occur at scales up to two orders of magnitude larger than the expanded uncertainties calculated in this work. While the effects of surface roughness and geometrical errors encountered in the XCT measurement of AM components may overshadow the variations shown here, the presented results provide a baseline for future experimentation to study AM specific effects. Future

work will investigate the assumption that the bias can be corrected for these measurements and will examine the transfer of uncertainty from this presented reference object to AM geometries which uphold the strict similarity conditions required by VDI 2630 Part 2.1. Further expansion of this idea to increasingly complex lattice geometries will also be investigated, as will the role of XCT imaging artifacts on measurement uncertainty and process repeatability.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 7

End cap measurements

	D_b	Lp_b	L_b	L_m	D_t	Lp_t	L_t
Value (mm)	3.9967	1.7241	2.0041	1.9402	3.9968	1.8141	1.9745
u_{cal} (μm)	0.4	0.4	0.4	0.4	0.4	0.4	0.4

Table 8

Pin location measurements

Measurement	L_1	L_2	L_3	L_4	L_5	L_6	L_7	L_8	L_9
Value (mm)	1.3402	0.9000	1.3623	0.9833	0.0793	1.0170	1.4493	1.0927	1.4801
u_{cal} (μm)	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4

Table 9

Pin diameter measurements

Measurement	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9
Value (mm)	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
u_{cal} (μm)	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5

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